Supplementary Notes for Security Self-Ablation

1 Experimental Setup

The security self-ablations isolate AMM pricing and modeled on-chain payment enforcement under adversarial participation. Each run uses 100 sellers, 200 buyers, 100 liquidity providers, and 30 tokens per resource over 100 epochs. Results are averaged over ten paired seeds (42–51), with malicious-participant ratios from 0% to 30%. Paired variants share physical populations, initial buyer endowments, and potential attack draws. This is an independent component experiment that retains the original security-model settings; it does not use the modified pricing curve of the granularity experiment.

In seller collusion, malicious sellers raise asks by 1.2–1.5× and reduce their listing rate to 30%, and full RWA is compared with RWA without AMM. In buyer collusion, malicious buyers lower bids to 60–80% and reduce spending per step to 30%, under the same AMM switch. In payment default, malicious buyers attempt nonpayment with probability 0.5; payment enforcement is switched on or off while AMM remains enabled. The *paid allocation ratio* is the physical capacity delivered with payment settled divided by initially available physical capacity. Unpaid delivery is excluded from its numerator.

2 Results and Analysis

As shown in `ablation_robustness.pdf`, removing AMM under seller collusion lowers paid allocation to 83.28% at a 30% malicious-seller ratio, compared with approximately 100% for full RWA. Inflated asks can restrict order-book trades, while the reduced listing rate slows the release of backed inventory. AMM quotes follow pool reserves rather than submitted asks, reducing direct exposure to the tested quotation manipulation. This explains the comparative pricing resilience observed under the modeled release conditions.

Under buyer collusion, the corresponding allocation without AMM is 96.28%, while full RWA remains at 100%. Without AMM, bids must cover asks and trades execute at their midpoint; with AMM, submitted bid shading cannot directly set the execution price. Both variants share the spending-rate reduction. Their aggregate paid-allocation outcomes therefore assess the pricing mechanism's sensitivity to this attack.

Under payment default, disabling enforcement lowers paid allocation to 83.86%, compared with 100% for the complete configuration. Enforcement prelocks 10% of initial buyer cash, recovers a defaulter's payment from collateral, applies a 10% trade-value penalty, and excludes the defaulter. Insufficient collateral rejects delivery and forfeits the available collateral. Consequently, unpaid consumption is prevented under these modeled rules.

Together, the comparisons identify complementary roles for reserve-based pricing and payment-settlement protection. Detection and collateral execution are simulation assumptions; these results do not constitute a live smart-contract or BFT-consensus benchmark.